BIRZEIT UNIVERSITY

COMP233 Discrete Mathematics

Quiz 1 – Chapters 2 and 3 – Section 4 – Answer Key and Marking Guide

100 points | 3 questions | Time: 30 minutes

Accept any logically equivalent form unless the item says otherwise. Accept ~ or ∼ for ¬, and ∴ / "therefore" for conclusions. Totals: Q1 = 20, Q2 = 35, Q3 = 45.

Question 1 [20 pts]

Let p = "The battery is charged", q = "The device turns on", and r = "The device is faulty."

a. Formalise each statement in symbols: [10 pts; 5 each]

(i) A charged battery is sufficient for the device to turn on.

(ii) Neither is the battery charged nor does the device turn on.

b. For the proposition "If the battery is charged and the device does not turn on, then the device is faulty", write its contrapositive and its negation in symbols. In your final answers, ¬ should apply only to single variables. [10 pts; 5 each]

a. (i) $p \to q$ (equivalently $\neg q \to \neg p$). "p is sufficient for q" means $p \to q$.

5 pts. 0 pts for $q \to p$ (direction reversed, which would be "necessary"). Full marks for any equivalent form.

a. (ii) $\neg p \land \neg q$ (equivalently $\neg(p \lor q)$).

5 pts. 2 pts for $\neg p \lor \neg q$ or $\neg(p \land q)$; 0 pts for other connectives.

b. Statement: $(p \land \neg q) \to r$.

Contrapositive: $\neg r \to \neg(p \land \neg q) \equiv \neg r \to (\neg p \lor q)$.

5 pts: 3 for the correct contrapositive structure $\neg r \to \neg(p \land \neg q)$, 2 for applying De Morgan to reach $\neg p \lor q$. 0 pts for the converse or inverse.

Negation: $\neg((p \land \neg q) \to r) \equiv (p \land \neg q) \land \neg r \equiv p \land \neg q \land \neg r$.

5 pts: 2 for $\neg((p \land \neg q) \to r)$ without expansion; full marks for $p \land \neg q \land \neg r$. 0 pts if the negation is written as a conditional.

Question 2 [35 pts]

a. Use a complete truth table to check whether $p \to (q \lor r)$ and $(p \land \neg q) \to r$ are logically equivalent. Explain your conclusion. [15 pts]

b. Simplify $(p \to q) \land (p \to \neg q)$ using logical equivalences. Name the law used at each step. Do not use a truth table. [10 pts]

c. From the premises $p \to q$, $q \to r$, $\neg r$, and $p \lor s$, derive s using rules of inference. Give the rule and the supporting premises or steps for each derived line. [10 pts]

a. Truth table (three variables, eight rows):

p	**q**	**r**	**q ∨ r**	**p → (q ∨ r)**	**¬q**	**p ∧ ¬q**	**(p ∧ ¬q) → r**
T	T	T	T	T	F	F	T
T	T	F	T	T	F	F	T
T	F	T	T	T	T	T	T
T	F	F	F	F	T	T	F
F	T	T	T	T	F	F	T
F	T	F	T	T	F	F	T
F	F	T	T	T	T	F	T
F	F	F	F	T	T	F	T

The columns for $p \to (q \vee r)$ and $(p \wedge \neg q) \to r$ agree in every row (both are F only in the row p = T, q = F, r = F, and T elsewhere), so the two statement forms are logically equivalent.

15 pts: 2 for all eight rows of truth-value combinations; 3 for the columns $q \vee r$ and $p \to (q \vee r)$; 3 for the columns $\neg q$ and $p \wedge \neg q$; 3 for the column $(p \wedge \neg q) \to r$; 4 for the conclusion stating equivalence because the two final columns are identical in every row. Deduct 1 pt per wrong cell within a column block (minimum 0 for that block). A table with only 4 rows earns at most 5 pts in total. A conclusion with no comparison of the columns earns at most 2 of the 4 pts. If the table is missing entirely: 0 pts.

b. Simplification:

$(p \to q) \wedge (p \to \neg q)$	
$\equiv (\neg p \vee q) \wedge (\neg p \vee \neg q)$	implication law ($p \to q \equiv \neg p \vee q$), applied twice
$\equiv \neg p \vee (q \wedge \neg q)$	distributive law
$\equiv \neg p \vee c$	negation law ($q \wedge \neg q \equiv c$)
$\equiv \neg p$	identity law ($\neg p \vee c \equiv \neg p$)

Final answer: $(p \to q) \wedge (p \to \neg q) \equiv \neg p$.

10 pts: 3 for the implication law step, 3 for the distributive step, 2 for the negation law, 2 for the identity law. Deduct 1 pt per step whose law is not named (max −3). Accept other correct orders of equivalent steps. 0 pts if a truth table is used instead.

c. One valid derivation (others are acceptable):

1.	$p \to q$	premise
2.	$q \to r$	premise
3.	$\neg r$	premise
4.	$p \vee s$	premise
5.	$\neg q$	modus tollens, 2 and 3
6.	$\neg p$	modus tollens, 1 and 5
7.	s	elimination (disjunctive syllogism), 4 and 6

Alternative route: 5. $p \to r$ by transitivity (hypothetical syllogism) from 1 and 2; 6. $\neg p$ by modus tollens from 5 and 3; 7. s by elimination from 4 and 6.

10 pts: 3 for deriving $\neg q$ (or $p \to r$), 3 for deriving $\neg p$, 4 for deriving s. Each derived line must name the rule and the lines it uses; deduct 1 pt per line with a missing or wrong rule name or missing references. 0 pts for a truth-table argument or for asserting s without steps.

Question 3 [45 pts]

a. The domain is all employees of a company. Let M(x) = "x is a manager," T(x) = "x has completed the security training," and K(x) = "x holds a key card." Consider:

"Every manager who has completed the security training holds a key card."

Write in symbols: (i) the statement, (ii) its contrapositive, (iii) its negation, and also give the negation as an English sentence. ¬ should apply only to predicates. [15 pts; 5 each]

For parts (b)–(d), let S be a set of students and C a set of courses, and let T(x, y) mean "student x has taken course y."

b. Formalise, using quantifiers with the specified domains: [10 pts; 5 each]

(i) Every course has been taken by at least one student.

(ii) Some student has taken every course.

c. Write the negation of statement (ii) in symbols, with ¬ moved inside so that it applies only to T(x, y). Then state the negation as an English sentence. [10 pts]

d. Let S = {Ali, Dana, Noor} and C = {Logic, Algebra}. The only courses taken are: Ali took Logic, Dana took Algebra, and Noor took Logic. Determine the truth value of statements (i) and (ii) from part (b), and justify each. What does this example show about whether (i) implies (ii)? [10 pts]

a. (i) $\forall x, (M(x) \wedge T(x)) \rightarrow K(x)$.

5 pts. 2 pts if $\wedge$ is used instead of $\rightarrow$ ($\forall x, M(x) \wedge T(x) \wedge K(x)$); 3 pts if the hypothesis omits one of $M(x)$, $T(x)$.

a. (ii) $\forall x, \neg K(x) \rightarrow \neg(M(x) \wedge T(x)) \equiv \forall x, \neg K(x) \rightarrow (\neg M(x) \vee \neg T(x))$.

5 pts: 3 for the correct structure, 2 for De Morgan giving $\neg M(x) \vee \neg T(x)$. 0 pts for the converse or inverse.

a. (iii) $\exists x, M(x) \wedge T(x) \wedge \neg K(x)$.

English: "There is a manager who has completed the security training but does not hold a key card." (Equivalently: some manager who completed the training has no key card.)

5 pts: 3 for the symbolic negation (must be $\exists$ with a conjunction; 1 pt if written $\neg\forall x, \ldots$ without moving the negation inside), 2 for a correct English sentence. 0 pts for the symbolic part if it contains $\rightarrow$.

b. (i) $\forall y \in C, \exists x \in S, T(x, y)$.

b. (ii) $\exists x \in S, \forall y \in C, T(x, y)$.

5 pts each. Quantifier order swapped: 2 pts. Domains missing or wrong (e.g., x ranging over C): deduct 1 pt. Both answers identical: at most 5 pts for the pair.

c. $\neg(\exists x \in S, \forall y \in C, T(x, y)) \equiv \forall x \in S, \exists y \in C, \neg T(x, y)$.

English: "For every student there is at least one course that the student has not taken." (Equivalently: no student has taken every course.)

10 pts: 6 for the symbolic form (each quantifier flipped and $\neg$ on $T(x, y)$ only; 2 pts if written as $\neg\exists x \in S, \forall y \in C, T(x, y)$ without moving $\neg$ inside; 3 pts if one quantifier is not flipped), 4 for the English sentence. The sentence must express "for every student, some course not taken"; "no student has taken any course" earns 0 for the English part.

d. Assignments: Ali–Logic, Dana–Algebra, Noor–Logic.

(i) True. Each course has at least one student: Logic was taken by Ali (and Noor); Algebra was taken by Dana.

(ii) False. No student has taken both courses: Ali and Noor have not taken Algebra, and Dana has not taken Logic. Equivalently, the negation from part (c) holds: for every student there is a course not taken.

Conclusion: this example makes (i) true and (ii) false, so (i) does not imply (ii). The order of $\forall$ and $\exists$ matters: $\exists x\, \forall y$ is stronger than $\forall y\, \exists x$.

10 pts: (i) 4 pts = 1 for True + 3 for naming a student for each course; (ii) 4 pts = 1 for False + 3 for showing every student misses some course (or that no student has both); 2 pts for stating that (i) does not imply (ii). If the student's own formalisations in (b) were wrong but the evaluation is consistent with them, award at most 2 pts per part.

Summary of marks: Q1a(i) 5 · Q1a(ii) 5 · Q1b 10 · Q2a 15 · Q2b 10 · Q2c 10 · Q3a 15 · Q3b 10 · Q3c 10 · Q3d 10. Total 100.